S.Y. B. Tech. Academic Year 2026-27 Semester: III

CSE20350/AID20210/CLD 20020: Data Structures Laboratory

WRITE UP

Experiment Number: 07

TITLE: Expression Evaluation using stack

PROBLEM STATEMENT:

Write a C program to evaluate postfix expression using stack.

OBJECTIVE:

1. To study Stack and its operations
2. To study the importance of expression evaluation

THEORY:

Expression evaluation is the process of calculating the value of an arithmetic expression. A stack is the most suitable data structure because it follows the Last In First Out (LIFO) principle, which matches the order in which operators and operands are processed.

Expression evaluation is commonly performed on:

- Postfix (Reverse Polish Notation)
- Prefix (Polish Notation)

Infix expressions are usually converted to postfix or prefix before evaluation.

Example:

Evaluate the expression: 3 5 + 4 *

Symbol	Operation	Stack
3	Push 3	3
5	Push 5	5, 3
+	Pop 5, pop 3 perform 3+5, push 8	8

4	Push 4	4, 8
*	Pop 4, pop 8 perform 4*8 push 32	32

IMPLEMENTATION:

- **PLATFORM:**
  - 64-bit Open source Linux or its derivatives.
  - Open Source C Programming tool like gcc/Eclipse Editor

- **TEST CONDITIONS:-**

1. Input postfix expression
2. Evaluate the value of expression

- **PSEUDO CODE:** ***//To be Written by Students***

  Write pseudo code to evaluate a postfix expression using stack

- **TIME COMPLEXITY:** ***//To be Written by Students***

  Write time complexity of above operations

- **CONCLUSION:**

  Thus, implemented Stack Operations to evaluate the value of an expression.

- **FAQs** ***//To be Written by Students***

  - Explain how postfix/prefix expression is evaluated.
  - What is ISP and ICP?
  - Give various applications of stack